

Routing & Layout Shell

▼ Relevant source files

Frontend/STUDER/src/app/app.routes.ts

Frontend/STUDER/src/app/shared/components/rich-text/entity-popover/entity-popover.component.ts

Frontend/STUDER/src/app/shared/components/sidebar/sidebar.component.css

Frontend/STUDER/src/app/shared/components/sidebar/sidebar.component.html

Frontend/STUDER/src/app/shared/components/sidebar/sidebar.component.ts

Frontend/STUDER/src/styles.css

The STUDER frontend utilizes a centralized routing configuration and a persistent layout shell pattern. This architecture separates public-facing landing pages from protected application features, which are wrapped in a consistent UI containing a global header and sidebar.

Route Definitions

The application's navigation is defined in `app.routes.ts` using Angular's `Routes` type. The structure is divided into three main categories: public routes, authenticated routes (guarded), and fallback redirects `Frontend/STUDER/src/app/app.routes.ts` 15-37

Public and Utility Routes

These routes are accessible without authentication or serve specific testing purposes:

- **Landing Page:** The root path `''` loads the `LandingComponent`

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- **Login/Register:** Accessible via `/login`, utilizing the `LoginRegisterComponent`

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- **Test Environment:** A `/test` route for `TestComponent`

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Authenticated Layout Group

The core application features are defined as children of the `LayoutComponent`. This group is protected by the `authGuard`, ensuring that only authenticated users can access the internal shell

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Path	Component	Description
/home	HomeComponent	Main dashboard view Frontend/STUDER/src/app/app.routes.ts 24
/discussions	DiscussionListComponent	List of public and joined discussions Frontend/STUDER/src/app/app.routes.ts 25
/discussions/:id	DiscussionDetailComponent	Threaded view of a specific discussion Frontend/STUDER/src/app/app.routes.ts 26
/messages	MessagesComponent	Direct messaging interface Frontend/STUDER/src/app/app.routes.ts 29
/search	SearchComponent	Global search results Frontend/STUDER/src/app/app.routes.ts 30
/account	(Redirect)	Redirects to user/me Frontend/STUDER/src/app/app.routes.ts 31
/user/:username	UserProfileComponent	Dynamic profile viewing for specific users Frontend/STUDER/src/app/app.routes.ts 33

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Layout Shell Implementation

The `LayoutComponent` serves as the primary structural wrapper for the application. It orchestrates the visibility of the `SidebarComponent` and the `HeaderComponent` (managed via `studer-header`).

Sidebar Navigation

The `SidebarComponent` provides the primary navigation links. It uses a data-driven approach with a `navItems` array to render navigation links dynamically

Navigation Item Structure

Each `NavItem` object contains:

- `label` : A translation key (e.g., `sidebar.home`)
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- `icon` : FontAwesome class names
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- `route` : The target URL path
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- `colorClass` : A specific CSS class used for theme-consistent hover and active states
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Visual States and Theming

The sidebar utilizes CSS variables defined in `styles.css` to manage its appearance across light and dark themes.

- **Active Link:** Tracked via `routerLinkActive="sidebar__link--active"`
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- **Color Mapping:** The sidebar assigns specific colors to different sections (e.g., Home is blue, Discussions is orange/warning) to match the STUDER branding
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Entity Popover Integration

A key feature of the layout shell is the ability to display contextual information through the `EntityPopoverComponent`. This component is triggered when users interact with rich-text entities (like `@usernames` or `#tags`) within the layout.

Data Resolution Flow

When an entity is hovered, the `EntityPopoverComponent` attempts to resolve data:

1. **Cache Check:** It queries the `EntityCacheService` using the entity value

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2. **API Fetch:** If not cached, it calls the relevant service (e.g., `UserService.getByUsername()`)

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3. **State Update:** Once data is retrieved, it updates the `popoverData` and triggers change detection via `ChangeDetectorRef.markForCheck()`

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Technical Diagrams

Route Architecture & Guards

This diagram illustrates the relationship between the route definitions and the authentication guards.

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Layout Component Interaction

This diagram bridges the visual layout shell with the underlying code entities and CSS variables.

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Styling and Global Utilities

The layout shell relies on a robust set of CSS variables and utility classes defined in `styles.css`.

Category	Variable / Class	Purpose
Theming	<code>--color-bg</code> , <code>--color-text-prim</code>	Swapped between <code>body.theme-light</code> and <code>body.theme-dark</code> Frontend/STUDER/src/styles.css 125-126
Layout	<code>--sidebar-width</code> , <code>--header-height</code>	Standardizes dimensions for fixed positioning Frontend/STUDER/src/styles.css 67-72
Utilities	<code>.username-link</code>	Standardized styling for user profile links across the app Frontend/STUDER/src/styles.css 161-167
Entities	<code>.entity--user</code> , <code>.entity--tag</code>	Color-coding for rich text entities in the UI Frontend/STUDER/src/styles.css 192-198
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